

Circuit Simulation Lab:10

D Flip-Flop

The purpose of this experiment is to introduce the design of simple sequential circuits, in this case D Flip Flop. A D (or Delay) Flip Flop is a digital electronic circuit used to delay the change of state of its output signal (Q) until the next rising edge of a clock timing input signal occurs.

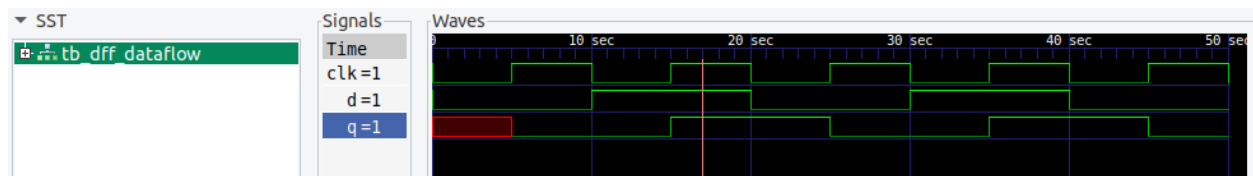
Describe the above in Verilog HDL and capture the Waveforms

- 1) Dataflow modeling
- 2) Behavior modeling
- 3) Structural modeling (Use the gates for designing the Flop)

Code:

Dataflow Modeling

```
module dff_dataflow(  
    input d,  
    input clk,  
    output reg q  
);  
  
    always @(posedge clk)  
    begin  
        q <= d;  
    end  
  
Endmodule
```

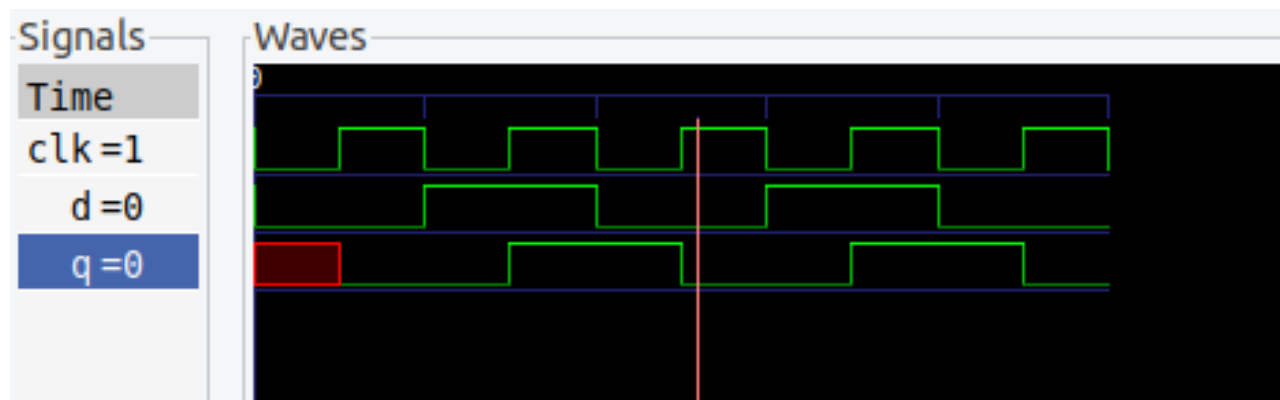


Behavioral Modeling

```
module dff_behavioral(  
  input d,  
  input clk,  
  output reg q  
);
```

```
  always @(posedge clk)  
  begin  
    q <= d;  
  end
```

```
Endmodule
```



Structural Modeling

```
module dff_structural(  
  input d,  
  input clk,  
  output q,  
  output qb  
);
```

```
  wire nd;  
  wire s,r;
```

```
  not g1(nd,d);
```

```
  and g2(s,d,clk);  
  and g3(r,nd,clk);
```

```
  nor g4(q ,r,qb);  
  nor g5(qb,s,q);
```

Endmodule

